



Principles of Inheritance and Variation

Unit VII GENETICS AND EVOLUTION

The work of **Mendel** and others who followed him gave us an idea of inheritance patterns.

The entire body of **molecular biology** was a consequent development with major contributions from **Watson, Crick, Nirenberg, Khorana, Kornberg's (father and son), Benzer, Monod, Brenner**, etc.

Profile JAMES WATSON

- James Dewey Watson was born in Chicago on 6 April 1928.
- In 1947, he received B.Sc. degree in Zoology.
- During these years his interest in bird-watching had matured into a serious desire to learn genetics.
- This became possible when he received a Fellowship for graduate study in Zoology at Indiana University, Bloomington, where he received his Ph.D degree in 1950 on a study of the effect of hard X-rays on bacteriophage multiplication.
- He met Crick and discovered their common interest in solving the DNA structure.
- Their first serious effort, was unsatisfactory.
- Their second effort based upon more experimental evidence and better appreciation of the nucleic acid literature resulted, early in March 1953, in the proposal of the complementary double-helical configuration.

Profile FRANCIS CRICK

- Francis Harry Compton Crick was born on 8 June 1916, at Northampton, England.
- He studied physics at University College, London and obtained a B.Sc. in 1937.
- He completed Ph.D. in 1954 on a thesis entitled X-ray Diffraction: Polypeptides and Proteins.
- A critical influence in Crick's career was his friendship with J.D. Watson, then a young man of 23, leading in 1953 to the proposal of the double-helical structure for DNA and the replication scheme.
- Crick was made an F.R.S. in 1959.

The honours to Watson with Crick include:

Year	Honour
1959	John Collins Warren Prize of the Massachusetts General Hospital
1960	Lasker Award
1962	Research Corporation Prize
1962	Nobel Prize

① [NOTE] The scientist portrait plates on source page 2 are text-only by the SS4.4 hard no - the photograph of a person is never embedded, converted or recreated; the name, dates, and achievements above carry everything examinable. The portrait captions as the source prints them are: "**JAMES WATSON**" and "**FRANCIS CRICK**".

Ch 4 PRINCIPLES OF INHERITANCE AND VARIATION - Chapter Opener

The chapter contents box lists six sections, ending at **4.6 Genetic Disorders**; the printed body, however, runs to **4.8**, adding **4.4 Polygenic Inheritance**, **4.5 Pleiotropy**, **4.6 Sex Determination**, **4.7 Mutation** and **4.8 Genetic Disorders**. The body numbering is authoritative for the chapter that follows.

- **Genetics** is the branch of biology that **deals with the inheritance, as well as the variation of characters from parents to offspring**. These and several related questions are dealt with, scientifically, in this branch.
- **Inheritance** is the process by which characters are passed on from parent to progeny; it is the **basis of heredity**.

- **Variation** is the degree by which progeny differ from their parents.

Humans knew from as early as 8000-1000 B.C. that one of the causes of variation was hidden in **sexual reproduction**.

- They **exploited the variations that were naturally present in the wild populations of plants and animals to selectively breed and select for organisms that possessed desirable characters.**
- For example, through **artificial selection and domestication from ancestral wild cows**, we have well-known Indian breeds, e.g., **Sahiwal cows in Punjab.**

One sentence the chapter summary adds and the body never does: Genetics is a branch of biology which deals with principles of inheritance and its practices.

And another: Progeny resembling the parents in morphological and physiological features has attracted the attention of many biologists.

And one more: Mendel was the first to study this phenomenon systematically.

Chapter 4 map	What it covers
4.1 Mendel's Laws of Inheritance	How Mendel set up the pea experiments and what he saw
4.2 Inheritance of One Gene	Monohybrid cross, Punnett square, testcross, Law of Dominance, Law of Segregation, incomplete dominance, co-dominance
4.3 Inheritance of Two Genes	Dihybrid cross, Law of Independent Assortment, chromosomal theory, linkage and recombination
4.4 Polygenic Inheritance	Traits that vary on a gradient rather than as two opposites (skin colour, height)
4.5 Pleiotropy	A single gene producing more than one phenotype
4.6 Sex Determination	XO, XY and ZW mechanisms; sex determination in humans and in honey bee
4.7 Mutation	How DNA sequences change, and what changes the genome
4.8 Genetic Disorders	Pedigree analysis, Mendelian disorders (colour blindness, haemophilia, sickle-cell anaemia, phenylketonuria, thalassemia), chromosomal disorders (Down's, Turner's, Klinefelter's)

☆ **[MEMORY AID - not in NCERT]** One chapter, three layers: **4.1-4.3** build Mendel's laws and tie them to chromosomes; **4.4-4.5** extend the one-gene, two-allele picture to many genes and many effects; **4.6-4.8** apply it - sex, mutation, and the disorders that fall out when any of it goes wrong. The Quick Recap at the end returns to this three-layer shape.

4.1

MENDEL'S LAWS OF INHERITANCE

It was during the **mid-nineteenth century** that headway was made in the understanding of inheritance.

- **Gregor Mendel, conducted hybridisation experiments on garden peas for seven years (1856-1863) and proposed the laws of inheritance in living organisms.**
- During Mendel's investigations into inheritance patterns it was for the first time that statistical analysis and mathematical logic were applied to problems in biology.
- His experiments had a large sampling size, which gave greater credibility to the data that he collected.
- Also, the confirmation of his inferences from experiments on successive generations of his test plants, proved that his results pointed to general rules of inheritance rather than being unsubstantiated ideas.

What Mendel worked with:

- Mendel investigated characters in the garden pea plant that were manifested as two opposing traits, e.g., tall or dwarf plants, yellow or green seeds.
- Mendel conducted such artificial pollination / cross pollination experiments using several true-breeding pea lines.
- **A true-breeding line is one that, having undergone continuous self-pollination, shows the stable trait inheritance and expression for several generations.**

- Mendel selected 14 true-breeding pea plant varieties, as pairs which were similar except for one character with contrasting traits.
- Some of the contrasting traits selected were: smooth or wrinkled seeds, yellow or green seeds, inflated (full) or constricted green or yellow pods and tall or dwarf plants (Figure 4.1, Table 4.1).

① [NOTE] Read the plate (Figure 4.1 labels). The plate shows seven pairs of contrasting traits in pea plant, each as a Character row with the Dominant trait and Recessive trait columns. The seven characters, in the order the plate prints them, are: Seed shape - Round / Wrinkled; Seed colour - Yellow / Green; Flower colour - Violet / White; Pod shape - Full / Constricted; Pod colour (no value carried in the label row); Flower position - Axial / Terminal; Stem height - Tall / Dwarf.

Table 4.1: Contrasting Traits Studied by Mendel in Pea

S.No.	Characters	Contrasting Traits
1.	Stem height	Tall / dwarf
2.	Flower colour	Violet / white (the source's own column heading prints Flouwer colour)
3.	Flower position	Axial / terminal
4.	Pod shape	Inflated / constricted
5.	Pod colour	Green / yellow
6.	Seed shape	Round / wrinkled
7.	Seed colour	Yellow / green

4.2

INHERITANCE OF ONE GENE

Take the example of **one such hybridisation experiment** carried out by Mendel, in which he **crossed tall and dwarf pea plants** to study the inheritance of **one gene** (Figure 4.2).

- He collected the seeds produced as a result of this cross and grew them to generate plants of the **first hybrid generation**.
- This generation is also called the **Filial₁ progeny or the F₁**.

① [NOTE] Read the plate (Figure 4.2 labels). The plate shows Steps in making a cross in pea: a Parent flower with the Petal lifted to reveal the Stigma, the Anther, the Stamen and the Carpel; the step Removal of anthers (Emasculation); and the step Transfer of pollen (Pollination), performed between two Parent plants.

- Mendel observed that all the F₁ progeny plants were tall, like one of its parents; none were dwarf (Figure 4.3).
- Mendel then self-pollinated the tall F₁ plants and to his surprise found that in the Filial₂ generation some of the offspring were 'dwarf'; the character that was not seen in the F₁ generation was now expressed.
- The proportion of plants that were dwarf were 1/4th of the F₂ plants while 3/4th of the F₂ plants were tall.
- The tall and dwarf traits were identical to their parental type and did not show any blending, that is all the offspring were either tall or dwarf, none were of in-between height (Figure 4.3).
- Similar results were obtained with the other traits that he studied: only one of the parental traits was expressed in the F₁ generation while at the F₂ stage both the traits were expressed in the proportion 3:1.
- The contrasting traits did not show any blending at either F₁ or F₂ stage.

① [NOTE] Read the plate (Figure 4.3 labels). The plate is a Diagrammatic representation of monohybrid cross. On the Parental line it marks a Tall plant and a Dwarf plant; the next line is the F₁ generation, all Tall; the Selfing arrow runs into the F₂ generation, where the Tall and Dwarf ratio is 3:1.

What the cross told Mendel:

- Based on these observations, Mendel proposed that something was being stably passed down, unchanged, from parent to offspring through the gametes, over successive generations.
- He called these things as 'factors'. Now we call them as genes.

- **Genes, therefore, are the units of inheritance.** They contain the information that is required to express a particular trait in an organism.
- **Genes which code for a pair of contrasting traits are known as alleles**, i.e., they are **slightly different forms of the same gene**.

Writing a cross in symbols:

- If we use alphabetical symbols for each gene, then the capital letter is used for the trait expressed at the F_1 stage and the small alphabet for the other trait.
- For example, in case of the character of height, T is used for the Tall trait and t for the 'dwarf', and T and t are alleles of each other.
- Hence, in plants the pair of alleles for height would be TT, Tt or tt.

Term	What it means	How the height alleles map to it
Genotype	The pair of alleles a plant actually carries.	TT (tall parent), Tt (F_1), tt (dwarf parent)
Phenotype	What the plant looks like.	Tall and dwarf
Homozygous	Both alleles in the pair are the same.	True-breeding parents are TT and tt
Heterozygous	The two alleles in the pair are different.	The F_1 is Tt
Dominant vs recessive	In a pair of dissimilar alleles, one is expressed in the phenotype and the other is hidden.	T is dominant over t , so Tt is phenotypically tall

- Mendel also proposed that in a true breeding, tall or dwarf pea variety the allelic pair of genes for height are identical or homozygous, TT and tt, respectively.
- TT and tt are called the genotype of the plant while the descriptive terms tall and dwarf are the phenotype.
- What then would be the phenotype of a plant that had a genotype Tt?
- As Mendel found the phenotype of the F_1 heterozygote Tt to be exactly like the TT parent in appearance, he proposed that in a pair of dissimilar factors, one dominates the other (as in the F_1) and hence is called the dominant factor while the other factor is recessive.
- In this case T (for tallness) is dominant over t (for dwarfness), that is recessive.
- (Do not use T for tall and d for dwarf because you will find it difficult to remember whether T and d are alleles of the same gene/character or not).
- Alleles can be similar as in the case of homozygotes TT and tt or can be dissimilar as in the case of the heterozygote Tt.
- Since the Tt plant is heterozygous for genes controlling one character (height), it is a monohybrid and the cross between TT and tt is a monohybrid cross.

From the F_1 and F_2 outcomes, Mendel inferred the segregation mechanism:

- From the observation that the recessive parental trait is expressed without any blending in the F_2 generation, we can infer that, when the tall and dwarf plant produce gametes, by the process of meiosis, the alleles of the parental pair separate or segregate from each other and only one allele is transmitted to a gamete.
- This segregation of alleles is a random process and so there is a 50 per cent chance of a gamete containing either allele, as has been verified by the results of the crossings.
- In this way the gametes of the tall TT plants have the allele T and the gametes of the dwarf tt plants have the allele t.
- During fertilisation the two alleles, T from one parent say, through the pollen, and t from the other parent, then through the egg, are united to produce zygotes that have one T allele and one t allele.

- The production of gametes by the parents, the formation of the zygotes, the F_1 and F_2 plants can be understood from a diagram called Punnett Square as shown in Figure 4.4.
- It was developed by a British geneticist, Reginald C. Punnett.
- It is a graphical representation to calculate the probability of all possible genotypes of offspring in a genetic cross.
- The possible gametes are written on two sides, usually the top row and left columns. All possible combinations are represented in boxes below in the squares, which generates a square output form.

First Punnett square: the parental cross and the F_1 :

	Tall parent gamete: T	Tall parent gamete: T
Dwarf parent gamete: t	Tt (F_1 , tall)	Tt (F_1 , tall)
Dwarf parent gamete: t	Tt (F_1 , tall)	Tt (F_1 , tall)

- The Punnett Square shows the parental tall TT (male) and dwarf tt (female) plants, the gametes produced by them and, the F_1 Tt progeny.
- The symbols for female and male are used to denote the female (eggs) and male (pollen) of the F_1 generation, respectively. (The printed glyphs are drawn symbols, not text - Pass 2 writes the words; the Unicode signs are banned by check 5.)
- The F_1 plant of genotype Tt when self-pollinated, produces gametes of the genotype T and t in equal proportion. When fertilisation takes place, the pollen grains of genotype T have a 50 per cent chance to pollinate eggs of the genotype T, as well as of genotype t. Also pollen grains of genotype t have a 50 per cent chance of pollinating eggs of genotype T, as well as of genotype t.

① [NOTE] Read the plate (Figure 4.4 labels). The plate is a Punnett square used to understand a typical monohybrid cross conducted by Mendel between true-breeding tall plants and true-breeding dwarf plants. The four corners carry the parental lines - a Tall plant with the TT genotype, a Dwarf plant with the tt genotype - with the Gametes row (a single T) and column (a single t). The middle of the grid is the F_1 generation - four cells, all Tt. Below the grid, the Selfing arrow leads to the F_2 generation, with two ratio lines: Phenotypic ratio: tall:dwarf 3:1 and Genotypic ratio: TT:Tt:tt 1:2:1.

Second Punnett square: the F_1 selfed to give the F_2 :

	F_1 gamete: T	F_1 gamete: t
F_1 gamete: T	TT (tall)	Tt (tall)
F_1 gamete: t	Tt (tall)	tt (dwarf)

- As a result of random fertilisation, the resultant zygotes can be of the genotypes TT, Tt or tt.
- From the Punnett square it is easily seen that 1/4th of the random fertilisations lead to TT, 1/2 lead to Tt and 1/4th to tt.
- Though the F_1 have a genotype of Tt, but the phenotypic character seen is 'tall'.
- At F_2 3/4th of the plants are tall, where some of them are TT while others are Tt.
- Externally it is not possible to distinguish between the plants with the genotypes TT and Tt.
- It is thus due to this dominance of one character over the other that all the F_1 are tall (though the genotype is Tt) and in the F_2 3/4th of the plants are tall (though genotypically 1/2 are Tt and only 1/4th are TT).
- This leads to a phenotypic ratio of 3/4 tall : (1/4 TT + 1/2 Tt) and 1/4 tt, i.e., a 3:1 ratio, but a genotypic ratio of 1:2:1.

4.2b

The same cross, written as a binomial expansion

- The $1/4 : 1/2 : 1/4$ ratio of TT:Tt:tt is mathematically condensable to the form of the binomial expression $(ax + by)^2$, that has the gametes bearing genes T or t in equal frequency of $1/2$.
- The expression is expanded as given below:

1

The gamete frequencies for a heterozygote Tt are $1/2$ T and $1/2$ t.

2

Squaring the gamete sum: $(1/2 T + 1/2 t)^2$.

3

Expanding: $1/4 TT + 1/2 Tt + 1/4 tt$.

- Mendel self-pollinated the F_2 plants and found that dwarf F_2 plants continued to generate dwarf plants in F_3 and F_4 generations.
- He concluded that the genotype of the dwarfs was homozygous-tt.
- From the preceding paragraphs it is clear that though the genotypic ratios can be calculated using mathematical probability, by simply looking at the phenotype of a dominant trait, it is not possible to know the genotypic composition.
- That is, for example, whether a tall plant from F_1 or F_2 has TT or Tt composition, cannot be predicted.

4.2c

Testcross - reading the genotype you cannot see

- Therefore, to determine the genotype of a tall plant at F_2 , Mendel crossed the tall plant from F_2 with a dwarf plant. This he called a testcross.
- In a typical testcross an organism (pea plants here) showing a dominant phenotype (and whose genotype is to be determined) is crossed with the recessive parent instead of self-crossing.
- The progenies of such a cross can easily be analysed to predict the genotype of the test organism.
- Figure 4.5 shows the results of typical testcross where violet colour flower (V) is dominant over white colour flower (v).

① [NOTE] Read the plate (Figure 4.5 labels). The plate is a **Diagrammatic representation of a test cross**. Both panels place a **Homozygous recessive tester (ww)** at the top. The other parent in each panel is the **Dominant Phenotype (Genotype unknown)**: left panel the unknown is labelled **WW** (homozygous dominant); right panel the unknown is labelled **Ww** (heterozygous). Left outcome: **Result - All flowers are violet, Interpretation - Unknown flower is homozygous dominant**. Right outcome: **Half of the flowers are violet and half of the flowers are white; Unknown flower is heterozygous**.

What the testcross distinguishes:

If the unknown dominant parent is...	Cross	Offspring
WW (homozygous dominant)	WW × ww	All Ww - all violet
Ww (heterozygous)	Ww × ww	Half Ww (violet), half ww (white)

☆ [MEMORY AID - not in NCERT] Testcross in one line: cross the unknown with the recessive; if any offspring show the recessive phenotype, the unknown was heterozygous. The recessive parent contributes only one kind of gamete, so every recessive offspring is a fingerprint of the dominant parent's hidden recessive allele.

4.2d

What the one-gene cross gives you: two laws

- Based on his observations on monohybrid crosses Mendel proposed two general rules to consolidate his understanding of inheritance in monohybrid crosses. Today these rules are called the **Principles or Laws of Inheritance**: the **First Law or Law of Dominance** and the **Second Law or Law of Segregation**.

- (i) Characters are controlled by discrete units called factors.
- (ii) Factors occur in pairs.
- (iii) In a dissimilar pair of factors one member of the pair dominates (dominant) the other (recessive).
- The law of dominance is used to explain the expression of only one of the parental characters in a monohybrid cross in the F_1 and the expression of both in the F_2 . It also explains the proportion of 3:1 obtained at the F_2 .

- This law is based on the fact that the alleles do not show any blending and that both the characters are recovered as such in the F_2 generation though one of these is not seen at the F_1 stage.
- Though the parents contain two alleles during gamete formation, the factors or alleles of a pair segregate from each other such that a gamete receives only one of the two factors.
- Of course, a homozygous parent produces all gametes that are similar while a heterozygous one produces two kinds of gametes each having one allele with equal proportion.

- When experiments on peas were repeated using other traits in other plants, it was found that sometimes the F_1 had a phenotype that did not resemble either of the two parents and was in between the two.
- The inheritance of flower colour in the dog flower (snapdragon or *Antirrhinum* sp.) is a good example to understand incomplete dominance.
- In a cross between true-breeding red-flowered (RR) and true-breeding white-flowered plants (rr), the F_1 (Rr) was pink (Figure 4.6).
- When the F_1 was self-pollinated the F_2 resulted in the following ratio 1 (RR) Red : 2 (Rr) Pink : 1 (rr) White.
- Here the genotype ratios were exactly as we would expect in any mendelian monohybrid cross, but the phenotype ratios had changed from the 3:1 dominant:recessive ratio.
- What happened was that R was not completely dominant over r and this made it possible to distinguish Rr as pink from RR (red) and rr (white).

① [NOTE] Read the plate (Figure 4.6 labels). The plate is the **Results of monohybrid cross in the plant Snapdragon**, where one allele is incompletely dominant over the other allele. The **P generation** row shows **Red (RR)** and **White (rr)**; the **Gametes** row carries **R** and **r**. The **F_1 generation** row reads **All pink (Rr)**. The **F_2 generation** grid is the four cells of a Punnett square - **RR, Rr, Rr, rr** - and the two ratio lines under it: **Phenotypic ratio: red:pink:white 1:2:1** and **Genotypic ratio: RR:Rr:rr 1:2:1**.

What the snapdragon result forces us to ask:

- Explanation of the concept of dominance: What exactly is dominance? Why are some alleles dominant and some recessive?
- To tackle these questions, we must understand what a gene does. Every gene, as you know by now, contains the information to express a particular trait.
- In a diploid organism, there are two copies of each gene, i.e., as a pair of alleles. Now, these two alleles need not always be identical, as in a heterozygote.
- One of them may be different due to some changes that it has undergone (about which you will read further on, and in the next chapter) which modifies the information that particular allele contains.
- Let's take an example of a gene that contains the information for producing an enzyme. Now there are two copies of this gene, the two allelic forms.
- Let us assume (as is more common) that the normal allele produces the normal enzyme that is needed for the transformation of a substrate S.
- Theoretically, the modified allele could be responsible for production of - (i) the normal/less efficient enzyme, or (ii) a non-functional enzyme, or (iii) no enzyme at all

- In the first case, the modified allele is equivalent to the unmodified allele, i.e., it will produce the same phenotype/trait, i.e., result in the transformation of substrate S. Such equivalent allele pairs are very common.
- But, if the allele produces a non-functional enzyme or no enzyme, the phenotype may be effected. The phenotype/trait will only be dependent on the functioning of the unmodified allele.
- The unmodified (functioning) allele, which represents the original phenotype is the dominant allele and the modified allele is generally the recessive allele.
- Hence, in the example above the recessive trait is seen due to non-functional enzyme or because no enzyme is produced.

4.2.2.2

Co-dominance

- Till now we were discussing crosses where the F_1 resembled either of the two parents (dominance) or was in-between (incomplete dominance). But, in the case of co-dominance the F_1 generation resembles both parents.
- A good example is different types of red blood cells that determine ABO blood grouping in human beings. ABO blood groups are controlled by the gene I.
- The plasma membrane of the red blood cells has sugar polymers that protrude from its surface and the kind of sugar is controlled by the gene.
- The gene (I) has three alleles I^A , I^B and i.
- The alleles I^A and I^B produce a slightly different form of the sugar while allele i does not produce any sugar.
- Because humans are diploid organisms, each person possesses any two of the three I gene alleles.
- I^A and I^B are completely dominant over i, in other words when I^A and i are present only I^A expresses (because i does not produce any sugar), and when I^B and i are present I^B expresses.
- But when I^A and I^B are present together they both express their own types of sugars: this is because of co-dominance. Hence red blood cells have both A and B types of sugars.
- Since there are three different alleles, there are six different combinations of these three alleles that are possible, and therefore, a total of six different genotypes of the human ABO blood types (Table 4.2).

Table 4.2: Table Showing the Genetic Basis of Blood Groups in Human Population

Allele from Parent 1	Allele from Parent 2	Genotype of offspring	Blood type of offspring
I^A	I^A	$I^A I^A$	A
I^A	I^B	$I^A I^B$	AB
I^A	i	$I^A i$	A
I^B	I^A	$I^A I^B$	AB
I^B	I^B	$I^B I^B$	B
I^B	i	$I^B i$	B
i	i	ii	O

- Do you realise that the example of ABO blood grouping also provides a good example of multiple alleles? Here you can see that there are more than two, i.e., three alleles, governing the same character.
- Since in an individual only two alleles can be present, multiple alleles can be found only when population studies are made.
- Occasionally, a single gene product may produce more than one effect.
- For example, starch synthesis in pea seeds is controlled by one gene. It has two alleles (B and b). Starch is synthesised effectively by BB homozygotes and therefore, large starch grains are produced. In contrast, bb homozygotes have lesser efficiency in starch synthesis and produce smaller starch grains.
- After maturation of the seeds, BB seeds are round and the bb seeds are wrinkled. Heterozygotes produce round seeds, and so B seems to be the dominant allele.
- But, the starch grains produced are of intermediate size in Bb seeds. So if starch grain size is considered as the phenotype, then from this angle, the alleles show incomplete dominance.

- Therefore, dominance is not an autonomous feature of a gene or the product that it has information for. It depends as much on the gene product and the production of a particular phenotype from this product as it does on the particular phenotype that we choose to examine, in case more than one phenotype is influenced by the same gene.

4.3

INHERITANCE OF TWO GENES

Mendel also worked with and crossed pea plants that differed in two characters, as is seen in the cross between a pea plant that has seeds with **yellow colour and round shape** and one that had seeds of **green colour and wrinkled shape** (Figure 4.7).

- Mendel found that the seeds resulting from the crossing of the parents, had yellow coloured and round shaped seeds.
- Thus, yellow colour was dominant over green and round shape dominant over wrinkled.
- Let us use the genotypic symbols Y for dominant yellow seed colour and y for recessive green seed colour, R for round shaped seeds and r for wrinkled seed shape.
- The genotype of the parents can then be written as RRYy and rryy.
- The cross between the two plants can be written down as in Figure 4.7 showing the genotypes of the parent plants.
- The gametes RY and ry unite on fertilisation to produce the F_1 hybrid RrYy.
- When Mendel self hybridised the F_1 plants he found that 3/4th of F_2 plants had yellow seeds and 1/4th had green. The yellow and green colours segregated in a 3:1 ratio. Round and wrinkled seed shape also segregated in a 3:1 ratio: just like in a monohybrid cross.

① [NOTE] Read the plate (Figure 4.7 labels). The plate is the **Results of a dihybrid cross where the two parents differed in two pairs of contrasting traits: seed colour and seed shape**. The **P generation** row carries **Round yellow** with **RR** and **YY**, and **Wrinkled green** with **rr** and **yy**. The **Gametes** row has **RY** and **ry**, uniting into the F_1 generation row of **RrYy**. The **Selfing** arrow runs to the dihybrid Punnett grid: **RRYY, RrYY, RRYy, rrYY, RRyy, Rryy, rrYy, rryy** in the F_2 generation; the F_1 gametes are the four-letter set **RY, Ry, rY, ry**. The ratio line under the grid: **Phenotypic ratio: round yellow : round green : wrinkled yellow : wrinkled green 9:3:3:1**.

4.3.1

Law of Independent Assortment

- In the dihybrid cross (Figure 4.7), the phenotypes round, yellow; wrinkled, yellow; round, green and wrinkled, green appeared in the ratio 9:3:3:1. Such a ratio was observed for several pairs of characters that Mendel studied.
- The ratio of 9:3:3:1 can be derived as a combination series of 3 yellow : 1 green, with 3 round : 1 wrinkled. This derivation can be written as follows:

1

(3 Round : 1 Wrinkled) (3 Yellow : 1 Green)

2

= 9 Round, Yellow : 3 Wrinkled, Yellow : 3 Round, Green : 1 Wrinkled, Green

- Based upon such observations on dihybrid crosses (crosses between plants differing in two traits) Mendel proposed a second set of generalisations that we call Mendel's Law of Independent Assortment. The law states that when two pairs of traits are combined in a hybrid, segregation of one pair of characters is independent of the other pair of characters.
- The Punnett square can be effectively used to understand the independent segregation of the two pairs of genes during meiosis and the production of eggs and pollen in the F_1 RrYy plant.
- Consider the segregation of one pair of genes R and r. Fifty per cent of the gametes have the gene R and the other 50 per cent have r. Now besides each gamete having either R or r, it should also have the allele Y or y.
- The important thing to remember here is that segregation of 50 per cent R and 50 per cent r is independent from the segregation of 50 per cent Y and 50 per cent y.
- Therefore, 50 per cent of the r bearing gametes has Y and the other 50 per cent has y. Similarly, 50 per cent of the R bearing gametes has Y and the other 50 per cent has y.

- Thus there are four genotypes of gametes (four types of pollen and four types of eggs).
- The four types are RY, Ry, rY and ry each with a frequency of 25 per cent or 1/4 of the total gametes produced.
- When you write down the four types of eggs and pollen on the two sides of a Punnett square it is very easy to derive the composition of the zygotes that give rise to the F₂ plants (Figure 4.7).

Activity (Figure 4.7 / 4.3.1): write down the 16 F₂ genotypes the dihybrid Punnett square yields (RRYY, RRYy, RRYy, RrYY, ... rryy) and group each under one of the four phenotypes: **Round Yellow, Round Green, Wrinkled Yellow, Wrinkled Green**. The four phenotype totals should land at **9 : 3 : 3 : 1**.

4.3.2

Chromosomal Theory of Inheritance

- Mendel published his work on inheritance of characters in 1865 but for several reasons, it remained unrecognised till 1900.
- Firstly, communication was not easy (as it is now) in those days and his work could not be widely publicised.
- Secondly, his concept of genes (or factors, in Mendel's words) as stable and discrete units that controlled the expression of traits and, of the pair of alleles which did not blend with each other, was not accepted by his contemporaries as an explanation for the apparently continuous variation seen in nature.
- Thirdly, Mendel's approach of using mathematics to explain biological phenomena was totally new and unacceptable to many of the biologists of his time.
- Finally, though Mendel's work suggested that factors (genes) were discrete units, he could not provide any physical proof for the existence of factors or say what they were made of.
- In 1900, three Scientists (de Vries, Correns and von Tschermak) independently rediscovered Mendel's results on the inheritance of characters.
- Also, by this time due to advancements in microscopy that were taking place, scientists were able to carefully observe cell division.
- This led to the discovery of structures in the nucleus that appeared to double and divide just before each cell division. These were called **chromosomes** (colored bodies, as they were visualised by staining).
- By 1902, the chromosome movement during meiosis had been worked out.
- Walter Sutton and Theodore Boveri noted that the behaviour of chromosomes was parallel to the behaviour of genes and used chromosome movement (Figure 4.8) to explain Mendel's laws (Table 4.3).

① [NOTE] Read the plate (Figure 4.8 labels). The plate is *Meiosis and germ cell formation in a cell with four chromosomes*. Can you see how chromosomes segregate when germ cells are formed? The plate is laid out in time order: a G₂ cell with paired chromosomes, a **Bivalent** stage (two homologous chromosomes paired), **Meiosis I** with its **anaphase** (the homologs pulled to opposite poles), **Meiosis II** (the sister chromatids separated), and the four final **Germ cells**.

Table 4.3: A Comparison between the Behaviour of Chromosomes and Genes

Chromosomes	Genes
Occur in pairs	Occur in pairs
Segregate at the time of gamete formation such that only one of each pair is transmitted to a gamete	Segregate at gamete formation and only one of each pair is transmitted to a gamete
Independent pairs segregate independently of each other	One pair segregates independently of another pair

- Recall that you have studied the behaviour of chromosomes during **mitosis** (equational division) and during **meiosis** (reduction division). The important things to remember are that chromosomes as well as genes occur in pairs.
- The two alleles of a gene pair are located on homologous sites on homologous chromosomes.
- Sutton and Boveri argued that the pairing and separation of a pair of chromosomes would lead to the segregation of a pair of factors they carried.

- Sutton united the knowledge of chromosomal segregation with Mendelian principles and called it the chromosomal theory of inheritance.
- Following this synthesis of ideas, experimental verification of the chromosomal theory of inheritance by Thomas Hunt Morgan and his colleagues, led to discovering the basis for the variation that sexual reproduction produced.
- Morgan worked with the tiny fruitflies, *Drosophila melanogaster* (Figure 4.10), which were found very suitable for such studies.
- They could be grown on simple synthetic medium in the laboratory.
- They complete their life cycle in about two weeks, and a single mating could produce a large number of progeny flies.
- Also, there was a clear differentiation of the sexes - the male and female flies are easily distinguishable.
- Also, it has many types of hereditary variations that can be seen with low power microscopes.

Independent assortment, the chromosome view (Figure 4.9):

- During Anaphase of meiosis I, the two chromosome pairs can align at the metaphase plate independently of each other (Figure 4.9).
- To understand this, compare the chromosomes of four different colour in the left and right columns. In the left column (Possibility I) orange and green is segregating together. But in the right hand column (Possibility II) the orange chromosome is segregating with the red chromosomes.

① [NOTE] Read the plate (Figure 4.8 labels). (Already given above - meiosis and germ cell formation in a cell with four chromosomes, laid out as G_2 to Bivalent to Meiosis I anaphase to Meiosis II to Germ cells.)

① [NOTE] Read the plate (Figure 4.9 labels). The plate is **Independent assortment of chromosomes**. The two halves of the plate are **Possibility I** and **Possibility II**; each half shows **Meiosis I - anaphase** at the top and **Meiosis II - anaphase** below, ending in four **Germ cells**. The four chromosomes are distinguished by colour - one long orange, one long yellow, one short red, one short green. In **Possibility I** the cell labelled **One long orange and short green chromosome and long yellow and short red chromosome at the same pole** is on the left, and the complementary arrangement on the right. In **Possibility II** the cell labelled **One long orange and short red chromosome and long yellow and short green chromosome at the same pole** is on the left, and the complementary arrangement on the right. The plate's meaning sits in those two arrangements - which two chromosomes end up in the same germ cell is not fixed.

① [NOTE] Read the plate (Figure 4.10 labels). The plate is ***Drosophila melanogaster* (a) Male (b) Female**. The plate carries no in-figure label rows beyond the title - it is a visual of the two sexes, with the male typically smaller and darker than the female. *Drosophila*'s short two-week life cycle, large brood size, and clear sex dimorphism (above) are what made it Morgan's model organism.

- Morgan carried out several dihybrid crosses in *Drosophila* to study genes that were sex-linked.
- The crosses were similar to the dihybrid crosses carried out by Mendel in peas. For example Morgan hybridised yellow-bodied, white-eyed females to brown-bodied, red-eyed males and intercrossed their F_1 progeny.
- He observed that the two genes did not segregate independently of each other and the F_2 ratio deviated very significantly from the 9:3:3:1 ratio (expected when the two genes are independent).
- Morgan and his group knew that the genes were located on the X chromosome (Section 4.4) and saw quickly that when the two genes in a dihybrid cross were situated on the same chromosome, the proportion of parental gene combinations were much higher than the non-parental type.
- Morgan attributed this due to the physical association or linkage of the two genes and coined the term linkage to describe this physical association of genes on a chromosome and the term recombination to describe the generation of non-parental gene combinations (Figure 4.11).
- Morgan and his group also found that even when genes were grouped on the same chromosome, some genes were very tightly linked (showed very low recombination) (Figure 4.11, Cross A) while others were

loosely linked (showed higher recombination) (Figure 4.11, Cross B).

- For example he found that the genes white and yellow were very tightly linked and showed only 1.3 per cent recombination while white and miniature wing showed 37.2 per cent recombination.
- His student Alfred Sturtevant used the frequency of recombination between gene pairs on the same chromosome as a measure of the distance between genes and 'mapped' their position on the chromosome.
- Today genetic maps are extensively used as a starting point in the sequencing of whole genomes as was done in the case of the Human Genome Sequencing Project, described later.

① [NOTE] Read the plate (Figure 4.11 labels). The plate is *Linkage: Results of two dihybrid crosses conducted by Morgan*. Cross A shows crossing between gene *y* and *w*; Cross B shows crossing between genes *w* and *m*. Here dominant wild type alleles are represented with (+) sign in superscript. Note: The strength of linkage between *y* and *w* is higher than *w* and *m*. The plate has two halves - Cross A on the left and Cross B on the right. Cross A's pair is **Yellow, white**; Cross B's pair is **White, miniature**. The **Wild type** parents sit at the top of each cross. The **F₁ generation** row gives the heterozygous flies; the next row is the **Gametes** each cross produces. Below the gametes, Cross A shows **Parental type (98.7%)** and **Recombinant types (1.3%)**; Cross B shows **Parental type (62.8%)** and **Recombinant types (37.2%)**. The phenotype labels in each cross are **yellow, white, yellow, white, miniature and wild type**.

4.4

POLYGENIC INHERITANCE

- Mendel's studies mainly described those traits that have distinct alternate forms such as flower colour which are either purple or white.
- But if you look around you will find that there are many traits which are not so distinct in their occurrence and are spread across a gradient.
- For example, in humans we don't just have tall or short people as two distinct alternatives but a whole range of possible heights.
- Such traits are generally controlled by three or more genes and are thus called as polygenic traits.
- Besides the involvement of multiple genes polygenic inheritance also takes into account the influence of environment.
- Human skin colour is another classic example for this.
- In a polygenic trait the phenotype reflects the contribution of each allele, i.e., the effect of each allele is additive.
- To understand this better let us assume that three genes A, B, C control skin colour in human with the dominant forms A, B and C responsible for dark skin colour and the recessive forms a, b and c for light skin colour.
- The genotype with all the dominant alleles (AABBCC) will have the darkest skin colour and that with all the recessive alleles (aabbcc) will have the lightest skin colour.
- As expected the genotype with three dominant alleles and three recessive alleles will have an intermediate skin colour. In this manner the number of each type of alleles in the genotype would determine the darkness or lightness of the skin in an individual.

4.5

PLEIOTROPY

- We have so far seen the effect of a gene on a single phenotype or trait.
- There are however instances where a single gene can exhibit multiple phenotypic expression. Such a gene is called a pleiotropic gene.
- The underlying mechanism of pleiotropy in most cases is the effect of a gene on metabolic pathways which contribute towards different phenotypes.
- An example of this is the disease phenylketonuria, which occurs in humans. The disease is caused by mutation in the gene that codes for the enzyme phenylalanine hydroxylase (single gene mutation). This manifests itself through phenotypic expression characterised by **mental retardation and a reduction in hair and skin pigmentation**.

- The mechanism of sex determination has always been a puzzle before the geneticists.
- The initial clue about the genetic/chromosomal mechanism of sex determination can be traced back to some of the experiments carried out in insects.
- In fact, the cytological observations made in a number of insects led to the development of the concept of genetic/chromosomal basis of sex-determination.
- Henking (1891) could trace a specific nuclear structure all through spermatogenesis in a few insects, and it was also observed by him that 50 per cent of the sperm received this structure after spermatogenesis, whereas the other 50 per cent sperm did not receive it.
- Henking gave a name to this structure as the x body but he could not explain its significance.
- Further investigations by other scientists led to the conclusion that the 'X body' of Henking was in fact a chromosome and that is why it was given the name X-chromosome.
- It was also observed that in a large number of insects the mechanism of sex determination is of the XO type, i.e., all eggs bear an additional X-chromosome besides the other chromosomes (autosomes).
- On the other hand, some of the sperms bear the X-chromosome whereas some do not.
- Eggs fertilised by sperm having an X-chromosome become females and, those fertilised by sperms that do not have an X-chromosome become males.
- Due to the involvement of the X-chromosome in the determination of sex, it was designated to be the sex chromosome, and the rest of the chromosomes were named as autosomes.
- Grasshopper is an example of XO type of sex determination in which the males have only one X-chromosome besides the autosomes, whereas females have a pair of X-chromosomes.
- In a number of other insects and mammals including man, XY type of sex determination is seen where both male and female have same number of chromosomes. Among the males an X-chromosome is present but its counterpart is distinctly smaller and called the Y-chromosome. Females, however, have a pair of X-chromosomes.
- Both males and females bear same number of autosomes. Hence, the males have autosomes plus XY, while female have autosomes plus XX.
- In human beings and in *Drosophila* the males have one X and one Y chromosome, whereas females have a pair of X-chromosomes besides autosomes (Figure 4.12a, b).
- In the above description you have studied about two types of sex determining mechanisms, i.e., XO type and XY type. But in both cases males produce two different types of gametes, (a) either with or without X-chromosome or (b) some gametes with X-chromosome and some with Y-chromosome. Such type of sex determination mechanism is designated to be the example of **male heterogamety**.
- In some other organisms, e.g., birds, a different mechanism of sex determination is observed (Figure 4.12 c). In this case the total number of chromosome is same in both males and females. But two different types of gametes in terms of the sex chromosomes, are produced by females, i.e., **female heterogamety**.
- In order to have a distinction with the mechanism of sex determination described earlier, the two different sex chromosomes of a female bird has been designated to be the Z and W chromosomes. In these organisms the females have one Z and one W chromosome, whereas males have a pair of Z-chromosomes besides the autosomes.

One sentence the chapter summary adds and the body never does: In chicken, sex chromosomes in male are ZZ, and in females are ZW.

① [NOTE] Read the plate (Figure 4.12 labels). The plate is **Determination of sex by chromosomal differences: (a, b) Both in humans and in *Drosophila* the female has a pair of XX chromosomes (homogametic) and the male XY (heterogametic) composition; (c) In many birds, female has a pair of dissimilar chromosomes Z_w and male two similar ZZ chromosomes.** The four labels the plate carries are XX (panel a and b, female), XY (panel a and b, male), ZW (panel c, female bird) and ZZ (panel c, male bird). The source prints the bird pair as Z_w and ZZ; the chapter's own paragraph just above names the female chromosome W (uppercase), and the conventional name is ZW - both spellings are kept as printed.

- It has already been mentioned that the sex determining mechanism in case of humans is XY type.
- Out of 23 pairs of chromosomes present, 22 pairs are exactly same in both males and females: these are the autosomes.
- A pair of X-chromosomes are present in the female, whereas the presence of an X and Y chromosome are determinant of the male characteristic.
- During spermatogenesis among males, two types of gametes are produced. 50 per cent of the total sperm produced carry the X-chromosome and the rest 50 per cent has Y-chromosome besides the autosomes.
- Females, however, produce only one type of ovum with an X-chromosome.
- There is an equal probability of fertilisation of the ovum with the sperm carrying either X or Y chromosome. In case the ovum fertilises with a sperm carrying X-chromosome the zygote develops into a female (XX) and the fertilisation of ovum with Y-chromosome carrying sperm results into a male offspring.
- Thus, it is evident that it is the genetic make up of the sperm that determines the sex of the child.
- It is also evident that in each pregnancy there is always 50 per cent probability of either a male or a female child.

- The sex determination in honey bee is based on the number of sets of chromosomes an individual receives.
- An offspring formed from the union of a sperm and an egg develops as a female (queen or worker), and an unfertilised egg develops as a male (drone) by means of parthenogenesis.
- This means that the males have half the number of chromosomes than that of a female. The females are diploid having 32 chromosomes and males are haploid, i.e., having 16 chromosomes.
- This is called as haplodiploid sex-determination system and has special characteristic features such as the males produce sperms by mitosis (Figure 4.13), they do not have father and thus cannot have sons, but have a grandfather and can have grandsons.

① [NOTE] Read the plate (Figure 4.13 labels - the honey bee plate, p21). The plate is Figure 4.13 Sex determination in honey bee (NCERT prints a SECOND Figure 4.13 on p22, the pedigree-symbol plate - they share the same printed number). The honey bee plate's labels: **Parents** at the top, with the two columns labelled **Female** and **Male**; the row below carries **Gametes**: (the female contributes 32, the male contributes 16); the middle row is the meiotic/mitotic branch with **Meiosis** on the female side and **Mitosis** on the male side; the bottom row is F_1 : - 32 on the female side (fertilised eggs) and 16 on the male side (unfertilised eggs). The 32 and 16 chromosome counts on the right are the diploid and haploid numbers.

- Mutation is a phenomenon which results in alteration of DNA sequences and consequently results in changes in the genotype and the phenotype of an organism.
- In addition to recombination, mutation is another phenomenon that leads to variation in DNA.
- As you will learn in Chapter 5, one DNA helix runs continuously from one end to the other in each chromatid, in a highly supercoiled form.
- Therefore loss (deletions) or gain (insertion/duplication) of a segment of DNA, result in alteration in chromosomes. Since genes are known to be located on chromosomes, alteration in chromosomes results in abnormalities or aberrations.
- Chromosomal aberrations are commonly observed in cancer cells.
- In addition to the above, mutation also arise due to change in a single base pair of DNA. This is known as point mutation. A classical example of such a mutation is sickle cell anemia.
- Deletions and insertions of base pairs of DNA, causes frame-shift mutations (see Chapter 5).
- However, there are many chemical and physical factors that induce mutations. These are referred to as mutagens. UV radiations can cause mutations in organisms - it is a mutagen.

- The idea that disorders are inherited has been prevailing in the human society since long. This was based on the heritability of certain characteristic features in families.
- After the rediscovery of Mendel's work the practice of analysing inheritance pattern of traits in human beings began.
- Since it is evident that control crosses that can be performed in pea plant or some other organisms, are not possible in case of human beings, study of the family history about inheritance of a particular trait provides an alternative.
- Such an analysis of traits in several of generations of a family is called the pedigree analysis. In the pedigree analysis the inheritance of a particular trait is represented in the family tree over generations.
- In human genetics, pedigree study provides a strong tool, which is utilised to trace the inheritance of a specific trait, abnormality or disease.
- Some of the important standard symbols used in the pedigree analysis have been shown in Figure 4.13.

① [NOTE] Read the plate (Figure 4.13 labels - the pedigree-symbol plate, p22). The plate is Figure 4.13 Symbols used in the human pedigree analysis (this is the second Figure 4.13 - same printed number, different plate from the honey bee one). The labels: Male, female, sex unspecified, affected individuals, mating, mating between relatives (consanguineous mating), parents above and children below (in order of birth - left to right), parents with male child affected with disease, and five unaffected offspring.

- DNA is the carrier of genetic information. It is hence transmitted from one generation to the other without any change or alteration.
- A number of disorders in human beings have been found to be associated with the inheritance of changed or altered genes or chromosomes.

- Broadly, genetic disorders may be grouped into two categories - Mendelian disorders and Chromosomal disorders.
- Mendelian disorders are mainly determined by alteration or mutation in the single gene.
- These disorders are transmitted to the offspring on the same lines as we have studied in the principle of inheritance. The pattern of inheritance of such Mendelian disorders can be traced in a family by the pedigree analysis.
- Most common and prevalent Mendelian disorders are Haemophilia, Cystic fibrosis, Sickle cell anaemia, Colour blindness, Phenylketonuria, Thalassemia, etc.
- By pedigree analysis one can easily understand whether the trait in question is dominant or recessive. Similarly, the trait may also be linked to the sex chromosome as in case of haemophilia.
- It is evident that this X-linked recessive trait shows transmission from carrier female to male progeny.

① [NOTE] Read the plate (Figure 4.14 labels). The plate is Representative pedigree analysis of (a) Autosomal dominant trait (for example: Myo tonic dystrophy) (b) Autosomal recessive trait (for example: Sickle-cell anaemia). The label row is empty in the source - the source's own column headers are the trait names - so the plate is read by what the source itself prints: a pedigree of the Myo tonic dystrophy family on the left (autosomal dominant) and a pedigree of the sickle-cell anaemia family on the right (autosomal recessive). The source spells *Myo tonic dystrophy* with the space inside the word; the conventional form is *myotonic dystrophy*. Both are kept as printed.

- **Colour Blindness:** It is a sex-linked recessive disorder due to defect in either red or green cone of eye resulting in failure to discriminate between red and green colour.
- This defect is due to mutation in certain genes present in the X chromosome.
- It occurs in about 8 per cent of males and only about 0.4 per cent of females.
- This is because the genes that lead to red-green colour blindness are on the X chromosome. Males have only one X chromosome and females have two.
- The son of a woman who carries the gene has a 50 per cent chance of being colour blind. The mother is not herself colour blind because the gene is recessive. That means that its effect is suppressed by her matching dominant normal gene.
- A daughter will not normally be colour blind, unless her mother is a carrier and her father is colour blind.

- **Haemophilia:** This sex linked recessive disease, which shows its transmission from unaffected carrier female to some of the male progeny has been widely studied.
- In this disease, a single protein that is a part of the cascade of proteins involved in the clotting of blood is affected.
- Due to this, in an affected individual a simple cut will result in non-stop bleeding.
- The heterozygous female (carrier) for haemophilia may transmit the disease to sons.
- The possibility of a female becoming a haemophilic is extremely rare because mother of such a female has to be at least carrier and the father should be haemophilic (unviable in the later stage of life).
- The family pedigree of Queen Victoria shows a number of haemophilic descendants as she was a carrier of the disease.

- **Sickle-cell anaemia:** This is an autosome linked recessive trait that can be transmitted from parents to the offspring when both the partners are carrier for the gene (or heterozygous).
- The disease is controlled by a single pair of allele, Hb^A and Hb^S .
- Out of the three possible genotypes only homozygous individuals for Hb^S ($Hb^S Hb^S$) show the diseased phenotype.
- Heterozygous ($Hb^A Hb^S$) individuals appear apparently unaffected but they are carrier of the disease as there is 50 per cent probability of transmission of the mutant gene to the progeny, thus exhibiting sickle-cell trait (Figure 4.15).
- The defect is caused by the substitution of Glutamic acid (Glu) by Valine (Val) at the sixth position of the beta globin chain of the haemoglobin molecule.
- The substitution of amino acid in the globin protein results due to the single base substitution at the sixth codon of the beta globin gene from GAG to GUG.
- The mutant haemoglobin molecule undergoes polymerisation under low oxygen tension causing the change in the shape of the RBC from biconcave disc to elongated sickle like structure (Figure 4.15).

① [NOTE] Read the plate (Figure 4.15 labels). The plate is Micro graph of the red blood cells and the amino acid composition of the relevant portion of beta-chain of haemoglobin: (a) From a normal individual; (b) From an individual with sickle-cell anaemia. The source spells it Micro graph. The plate labels (a) on the left and (b) on the right. The labels on the two gene cards are Normal Hb(A) gene and Sickle-cell Hb(S) gene. The mRNA lines are mRNA GAG and mRNA GUG. The peptide chains are HbA peptide and HbS peptide, and the amino acid sequences printed under them are Val His Leu Thr Pro Glu Glu (HbA) and Val His Leu Thr Pro Val Glu (HbS) - the difference is one residue: Glu at position 6 in HbA, Val at position 6 in HbS.

- **Phenylketonuria:** This inborn error of metabolism is also inherited as the autosomal recessive trait.
- **The affected individual lacks an enzyme that converts the amino acid phenylalanine into tyrosine.**
- **As a result of this phenylalanine is accumulated and converted into phenylpyruvic acid and other derivatives.**
- **Accumulation of these in brain results in mental retardation.**
- **These are also excreted through urine because of its poor absorption by kidney.**

- **Thalassemia:** This is also an autosome-linked recessive blood disease transmitted from parents to the offspring when both the partners are unaffected carrier for the gene (or heterozygous).
- **The defect could be due to either mutation or deletion which ultimately results in reduced rate of synthesis of one of the globin chains (alpha and beta chains) that make up haemoglobin.** This causes the formation of abnormal haemoglobin molecules resulting into anaemia which is characteristic of the disease.
- **Thalassemia can be classified according to which chain of the haemoglobin molecule is affected.** In alpha Thalassemia, production of alpha globin chain is affected while in beta Thalassemia, production of beta globin chain is affected.
- **alpha Thalassemia is controlled by two closely linked genes HBA1 and HBA2 on chromosome 16 of each parent and it is observed due to mutation or deletion of one or more of the four genes.** The more genes affected, the less alpha globin molecules produced.
- **While beta Thalassemia is controlled by a single gene HBB on chromosome 11 of each parent and occurs due to mutation of one or both the genes.**
- **Thalassemia differs from sickle-cell anaemia in that the former is a quantitative problem of synthesising too few globin molecules while the latter is a qualitative problem of synthesising an incorrectly functioning globin.**

- **The chromosomal disorders on the other hand are caused due to absence or excess or abnormal arrangement of one or more chromosomes.**
- **Failure of segregation of chromatids during cell division cycle results in the gain or loss of a chromosome(s), called aneuploidy.**
- **For example Down's syndrome results in the gain of extra copy of chromosome 21.**
- **Similarly, Turner's syndrome results due to loss of an X chromosome in human females.**
- **Failure of cytokinesis after telophase stage of cell division results in an increase in a whole set of chromosomes in an organism and, this phenomenon is known as polyploidy.** This condition is often seen in plants.
- **The total number of chromosomes in a normal human cell is 46 (23 pairs). Out of these 22 pairs are autosomes and one pair of chromosomes are sex chromosome.**
- **These situations are known as trisomy or monosomy of a chromosome, respectively.** Such a situation leads to very serious consequences in the individual.
- **Down's Syndrome, Turner's syndrome, Klinefelter's syndrome are common examples of chromosomal disorders.**

- **Down's Syndrome:** The cause of this genetic disorder is the presence of an additional copy of the chromosome number 21 (trisomy of 21).
- **This disorder was first described by Langdon Down (1866).**
- **The affected individual is short statured with small round head, furrowed tongue and partially open mouth (Figure 4.16).** Palm is broad with characteristic palm crease. Physical, psychomotor and mental development is retarded.

① [NOTE] Read the plate (Figure 4.16 labels). The plate is **A representative figure showing an individual inflicted with Down's syndrome and the corresponding chromosomes of the individual**. The label row of the plate is empty in the source, and the only fact-bearing labels NCERT prints on the plate are the six physical signs called out as: **Broad flat face, Flat back of head, Many 'loops' on finger tips, Palm crease, Big and wrinkled tongue, and Congenital heart disease.**

Klinefelter's Klinefelter's Syndrome (47, XXY)

- **Klinefelter's Syndrome:** This genetic disorder is also caused due to the presence of an additional copy of X chromosome resulting into a karyotype of **47, XXY**.
- **Such an individual has overall masculine development, however, the feminine development (development of breast, i.e., Gynaecomastia) is also expressed (Figure 4.17a).** Such individuals are sterile.

Turner's Turner's Syndrome (45, XO)

- **Turner's Syndrome:** Such a disorder is caused due to the absence of one of the X chromosomes, i.e., **45 with XO**, Such females are sterile as ovaries are rudimentary besides other features including lack of other secondary sexual characters (Figure 4.17b).

① [NOTE] Read the plate (Figure 4.17 labels). The plate is **Diagrammatic representation of genetic disorders due to sex chromosome composition in humans: (a) Klinefelter Syndrome; (b) Turner's Syndrome**. The source's own spelling is **represe-ntation**, hyphenated across a line break. The two label rows on the plate, one per panel, are: panel (a) **Tall stature with feminised character (development of breast, i.e., Gynaecomastia);** panel (b) **Short stature and under developed feminine character.**

One sentence the chapter summary adds and the body never does: These can be easily studied by analysis of Karyotypes.

Recap QUICK RECAP

- **Genetics** is a branch of biology which deals with **principles of inheritance and its practices** (summary-only wording; the body's definition reads differently).
- **Progeny resembling the parents in morphological and physiological features** has attracted the attention of many biologists (summary-only wording).
- **Mendel was the first to study this phenomenon systematically** (summary-only wording).
- **While studying the pattern of inheritance in pea plants of contrasting characters, Mendel proposed the principles of inheritance, which are today referred to as 'Mendel's Laws of Inheritance'.**
- **He proposed that the 'factors' (later named as genes) regulating the characters are found in pairs known as alleles.**
- **He observed that the expression of the characters in the offspring follow a definite pattern in different-first generations (F_1), second (F_2) and so on.**
- **Some characters are dominant over others.**
- **The dominant characters are expressed when factors are in heterozygous condition (Law of Dominance).**
- **The recessive characters are only expressed in homozygous conditions.**
- **The characters never blend in heterozygous condition.**
- **A recessive character that was not expressed in heterozygous conditon may be expressed again when it becomes homozygous.** (NCERT spelling **conditon** preserved.)
- **Hence, characters segregate while formation of gametes (Law of Segregation).**
- **Not all characters show true dominance. Some characters show incomplete, and some show co-dominance.**
- **When Mendel studied the inheritance of two characters together, it was found that the factors independently assort and combine in all permutations and combinations (Law of Independent Assortment).**
- **Different combinations of gametes are theoretically represented in a square tabular form known as 'Punnett Square'.**

- The factors (now known as gene) on chromosomes regulating the characters are called the **genotype** and the physical expression of the characters is called **phenotype**. (NCERT spelling chraracters preserved.)
- After knowing that the genes are located on the chromosomes, a good correlation was drawn between Mendel's laws: segregation and assortment of chromosomes during meiosis. The Mendel's laws were extended in the form of 'Chromosomal Theory of Inheritance'.
- Later, it was found that Mendel's law of independent assortment does not hold true for the genes that were located on the same chromosomes. These genes were called as **linked genes**.
- Closely located genes assorted together, and distantly located genes, due to recombination, assorted independently. Linkage maps, therefore, corresponded to arrangement of genes on a chromosome.
- Many genes were linked to sexes also, and called as **sex-linked genes**.
- The two sexes (male and female) were found to have a set of chromosomes which were common, and another set which was different. The chromosomes which were different in two sexes were named as **sex chromosomes**. The remaining set was named as **autosomes**.
- In humans, a normal female has 22 pairs of autosomes and a pair of sex chromosomes (Xx). A male has 22 pairs of autosomes and a pair of sex chromosome as XY. (NCERT spelling Xx preserved.)
- In chicken, sex chromosomes in male are ZZ, and in females are ZW. (Summary-only wording; the body's paragraph 4.6 names the chromosomes Z and W but does not name the chicken.)
- Mutation is defined as change in the genetic material.
- A point mutation is a change of a single base pair in DNA.
- Sickle-cell anemia is caused due to change of one base in the gene coding for beta-chain of hemoglobin.
- Inheritable mutations can be studied by generating a pedigree of a family.
- Some mutations involve changes in whole set of chromosomes (polyploidy) or change in a subset of chromosome number (aneuploidy).
- This helped in understanding the mutational basis of genetic disorders.
- Down's syndrome is due to trisomy of chromosome 21, where there is an extra copy of chromosome 21 and consequently the total number of chromosome becomes 47.
- In Turner's syndrome, one X chromosome is missing and the sex chromosome is as X0, and in Klinefelter's syndrome, the condition is XXY.
- These can be easily studied by analysis of **Karyotypes**. (Summary-only wording; the body gives 'a karyotype of 47, XXY' as a phrase, never 'analysis of Karyotypes' as a method.)

Appendix TERMS USED IN THE EXERCISES

The exercises at the end of this NCERT chapter assume four things the chapter itself never spells out. All four are worked out below, each visibly marked as an answer drawn from this chapter's own sentences and not as text carried in from elsewhere.

- 16 exercises, 4 answered by design (GAP), 12 unanswered by design (COVERED), 0 overlooked.

Q1 Advantages of selecting pea plant for experiment by Mendel

Worked-out answer, assembled from this chapter's own sentences in section 4.1 (not NCERT text):

- **True-breeding lines.** The pea plant has many **true-breeding pea lines** (F038) - varieties that, under continuous self-pollination, show the **stable trait inheritance and expression for several generations** (F039).
- **Characters with two opposing traits.** The pea plant has **characters ... that were manifested as two opposing traits** (F037) - tall/dwarf, yellow/green, round/wrinkled - 14 such pairs in total (F040).
- **Artificial cross-pollination is straightforward.** Mendel conducted **artificial pollination / cross pollination experiments** (F038) - emasculation and pollen transfer are simple to do on a pea flower (Figure 4.2).
- **Short life cycle and many offspring.** The 7-year window (1856-1863, F033) implies a manageable generation time and enough progeny to do the statistical analysis that the chapter credits Mendel with introducing for the first time in biology (F034).
- **Large sampling size, giving credibility to the data.** The chapter calls this out by name (F035).

☆ **[MEMORY AID - not in NCERT]** Five reasons, one mnemonic: **True-breeding, Two-opposing, Cross-ability, Short cycle, Sample size - T-T-C-S-S**. Or, said the other way: every property the body says the pea has is the reason a question calling them 'advantages' would want named.

Q3 Gamete types for a diploid heterozygous at 4 loci (2^n rule)

Worked-out answer, extension of the 4.3.1 statement (not NCERT text):

- Section 4.3.1 (F188) states that the F_1 plant $RrYy$ produces **four types of gametes (RY, Ry, rY, ry) each with a frequency of 25 per cent or 1/4**.
- Two heterozygous loci to $2^2 = 4$ gamete types, each at **1/4**.
- Generalising: n heterozygous loci to 2^n gamete types, each at **$1/2^n$** .
- For $n = 4$: $2^4 = 16$ gamete types, each at **1/16**.

☆ **[MEMORY AID - not in NCERT]** 2^n gametes, each at $1/2^n$: two loci to 4 at 1/4; three loci to 8 at 1/8; four loci to 16 at 1/16. The source's 4.3.1 gives the 2-locus case as the only worked example; the rest is reasoning on top of it.

Q6 Homozygous female x heterozygous male, single locus

Worked-out answer, drawing on the Law of Dominance rows of 4.2.1 (not NCERT text):

- Cross: **TT × Tt** (homozygous dominant female, heterozygous male).
- Female gametes: all **T**.
- Male gametes: **1/2 T** and **1/2 t**.
- Offspring: **1/2 TT** and **1/2 Tt**.
- **Genotypic ratio: 1:1**.
- **Phenotypic ratio: all tall (1:0)** - by the Law of Dominance (4.2.1), T is dominant over t, so both TT and Tt are phenotypically tall (F072, F074).

☆ **[MEMORY AID - not in NCERT]** **Homozygous × heterozygous, single locus: 1 TT : 1 Tt** genotypically, **all tall** phenotypically. Same gamete split as a testcross (1:1) but with a dominant parent instead of a recessive one, so no recessive phenotype appears.

Q7 Cross TtYy (tall, yellow) × Tt yy (tall, green)

Worked-out answer, derived from the 4.3.1 gamete-frequency rows (not NCERT text):

- Parent 1 (TtYy) gametes (4.3.1, F188): **1/4 TY, 1/4 Ty, 1/4 tY, 1/4 ty**.
- Parent 2 (Tt yy) gametes: **1/2 Ty** and **1/2 ty** (T/t segregates 1:1; only y is available on the colour locus).
- Combine: **4 × 2 = 8** equally-likely cells (each at **1/8**):

Parent 1 gamete	x Parent 2	Offspring	Phenotype
TY (1/4)	Ty (1/2)	TTYy (1/8)	Tall, yellow
TY (1/4)	ty (1/2)	TtYy (1/8)	Tall, yellow
Ty (1/4)	Ty (1/2)	TTyy (1/8)	Tall, green
Ty (1/4)	ty (1/2)	Ttyy (1/8)	Tall, green
tY (1/4)	Ty (1/2)	TtYy (1/8)	Tall, yellow
tY (1/4)	ty (1/2)	ttYy (1/8)	Dwarf, yellow
ty (1/4)	Ty (1/2)	Ttyy (1/8)	Tall, green
ty (1/4)	ty (1/2)	ttyy (1/8)	Dwarf, green

- **Phenotypic totals:** tall-yellow **3/8** (TTYy 1/8 + TtYy 1/8 + TtYy 1/8); tall-green **3/8** (TTyy 1/8 + Ttyy 1/8 + Ttyy 1/8); dwarf-yellow **1/8** (ttYy); dwarf-green **1/8** (ttyy).
- **(a) Tall and green: 3/8**.
- **(b) Dwarf and green: 1/8** - the t gametes from Parent 1 are required; without them the dwarf classes vanish, which is the error the corrected table removes.

☆ **[MEMORY AID - not in NCERT]** $TtYy \times Ttyy$ in one line: $3/8$ tall-yellow : $3/8$ tall-green : $1/8$ dwarf-yellow : $1/8$ dwarf-green. Height still segregates 3 tall : 1 dwarf (Parent 1 is Tt ; Parent 2 is Tt too), and colour segregates 1 yellow : 1 green (only Parent 1 carries Y). Independent assortment multiplies the two monohybrid ratios.